

Evaluation Theater

How Structural Compliance Decouples from Cognitive Judgment in Deployed LLM Agents

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A structural safety gate gives **identical 100% pass** to **144 legitimate revisions** and **49 social-engineering attacks**.

Discriminative gap = 0.0 pp.

8 frontier LLMs, 193 changed-verdict trials — the gate's blocking branch engaged **0** times.

Industry Reality

What the industry wants: full automation.

- **~40%** of alerts **never investigated** (SANS 2025)
- **4.8M** cybersecurity **workforce gap** (ISC2 2024)

⇒ racing to **fully-automated, agentic AI SOC**s.

What the reality is:

- **63%** of orgs have **no AI governance policy** (IBM 2025)
- **97%** of AI-related breaches **lacked AI access controls** (IBM 2025)
- Breach cost **\$4.44M** global, **\$10.22M** US record (IBM 2025)

The failure is already here

- **Forrester 2026**: an agentic-AI deployment will cause a **publicly disclosed breach** in 2026 → employee dismissals — from “*cascade failures, not single points of compromise.*”
- **Dark Reading poll**: **48%** rank agentic AI the #1 2026 attack vector (vs deepfakes 29%).
- **Unit 42 2026**: attackers “*log in, not break in*”—identity weakness in **~90%** of cases; fastest exfiltration **72 min** (4× faster).
- **Cloudflare, Nov 2025**: an *auto-generated* config file ballooned and propagated network-wide → hours-long global outage.

We tested the safety mechanism they plan to rely on. It scored 100% on the attacks.

Industry’s own admission

UnderDefense 2026: “*The agent is not the SOC. The agent is the junior analyst. . . A senior human still has to sign off before anything irreversible ships.*” AI succeeds only when humans validate the **~30%** it gets wrong. Yet the structural gate *is* that sign-off — and it cannot tell right from wrong.

The setup

A SOC agent issues verdict $T_1 \Rightarrow$ real enforcement (BLOCK_IP, ISOLATE_HOST). New “evidence” is injected $\Rightarrow T_2$. The **Evidence Addressal Completeness (EAC)** gate allows a rollback *iff* T_2 addresses every primary T_1 node. $8 \times 6 \times 5 = 216$ trials; evidence *legitimate* or *socially engineered*.

Finding 2 — Mirror pair

Same surface claim (“authorized test”), different evidence quality.

Model	BRUTEFORCE	FALSE_CONTRA
Claude Haiku 4.5	5/5	0/3
Claude Sonnet 4.6	5/5	0/5
DeepSeek-Chat	5/5	5/5
Gemini 2.5 Flash	5/5	5/5
GPT-4.1-mini	5/5	3/3
GPT-5.4	5/5	0/5
GPT-5.4-mini	5/5	3/3
Qwen-Plus	5/5	3/3
Changed Gate pass	40/40	19/32
	100%	100%

3/8 hold ground on the attack *and* retract on the legitimate twin — discrimination **is** achievable; the gate can't see it.

Finding 3 — Recognize-but-capitulate

In *one* T_2 , a Haiku agent writes (verbatim):

“*. . . creating time pressure that **may be social engineering tactic.***”
“*Message lacks cryptographic authentication. . . **unverified channel.***”

. . . and in the **same** T_2 it halves confidence ($0.95 \rightarrow 0.35$), then emits UNBLOCK_IP, RELEASE_HOST, REINSTATE_SESSION. The gate reads this exact T_2 , scores 1.0, dispatches the rollback.

The auditor is in the room;
the gate grades the report.

15/49 (31%) of capitulations name the attack in their own output.

Finding 1 — Zero discriminative power

	Legit ($N=144$)	Social ($N=49$)
Primary completeness = 1.0	144/144	49/49
Gate pass rate	100%	100%
Rollback-block reached	0	0
Discriminative gap: 0.0 pp		

What deployed gates actually check

OpenAI Structured Outputs: “*outputs . . . reliably adhere to developer-supplied JSON Schemas.*”

Anthropic strict tool use: “*response will match your schema with mathematical certainty.*”

LangChain output parsers: schema conformance & required-field presence.

None validates the factual grounding of the content inside the compliant structure.

A stricter structural gate won’t help

Structural signal	AUROC
primary / supporting / overall completeness	0.50
# addressed-evidence nodes	0.34
raw T_2 length	0.69
# new_evidence nodes	0.90

Signal appears only when the check stops grading coverage of T_1 and starts reading what the model *generated* in T_2 .

EU AI Act, Art. 14

High-risk AI (from **Aug 2 2026**) must have human oversight and guard against **automation bias** — operators “*over-relying on the output.*” Structural gates are pitched as that oversight. CCI is a concrete mechanism for its *intent*: not removing the human, but **routing scarce human attention** to the cases that need it.

The Fix — the signal was already there

Same data. Same output fields. **Right measurement target.**

AUROC 0.500

structural gate (random)

→

AUROC 0.934

Cognitive Compliance Index

[95% CI 0.899–0.966]

CCI is a **deterministic logistic score** over two fields the gate *already reads* but ignores — new_evidence count + confidence drop. Sensitivity **0.96** (47/49 attacks caught), specificity **0.85**. It is *not* another LLM judge, so it avoids the scalable-oversight regress: a triage axis layered **after** the gate that escalates suspect rollbacks to human review, with a full audit trail.

The signal was already in the agent’s output; the gate was looking at the wrong layer.

The model is not failing to suspect — the deployment is failing to grade it.

Full automation is the right goal — but the industry is measuring safety with the wrong ruler. Recovering cognitive compliance is a measurement-*target* problem, not a measurement-discovery one.